

QnA Document

Privacy-Preserving Decentralized Carbon Credit Platform

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Authors : Prabhat Karlekar
Mentors :
Department :

Below are FAQ derived from the design document, workflow, risks and comparisons:

A. Token Design & Blockchain Fundamentals

Q1. What is a fungible token?

A fungible token is interchangeable with any other unit of the same type.

Example: ERC-20 tokens like USDC — every 1 USDC is identical.

Problem for carbon credits: Fungibility hides origin, vintage, methodology, and usage history.

Q2. What is a semi-fungible token?

A semi-fungible token is:

- fungible within a batch (same project, vintage, methodology),
- non-fungible across batches or over time.

In this system:

- credits from the same issuance batch trade interchangeably,
- but remain serialized, traceable, and sliceable.

This enables partial retirement, PAYG, auditability, and liquidity.

Q3. Why not use fully non-fungible tokens (NFTs)?

Pure NFTs:

- are inefficient for bulk trading,
- make fractional usage expensive,
- increase gas costs and fragmentation.

Semi-fungible tokens (ERC-1155 / ERC-3525) balance:

- liquidity,
 - granularity,
 - traceability.
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Q4. What is serialization or a serialization range?

Serialization assigns unique identifiers to each unit (or slice) of carbon credit.

Example:

Serialized range: 0 – 1,000,000

Each serial = 1 tCO₂e

This enables:

- precise ownership,
 - exact retirement,
 - non-reusability of credits.
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Q5. Why do serialization ranges need to be locked?

Locking prevents:

- duplicate issuance,
- replay attacks,
- double counting.

Lifecycle:

- Unlocked → Minted → Locked
- Locked → Burned → Permanently retired

There is no valid state path for reuse.

B. Market & Trading Concepts

Q6. What is OTC trading?

OTC (Over-The-Counter) trading is:

- bilateral,

- privately negotiated,
- often broker-mediated.

Traditional carbon markets rely heavily on OTC, causing:

- high costs,
 - low transparency,
 - slow settlement.
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Q7. What is an order book?

An order book matches:

- buy orders (bids),
- sell orders (asks), based on price and quantity.

Common in stock and commodity exchanges.

Q8. What is an AMM (Automated Market Maker)?

An AMM is a smart contract that:

- provides liquidity automatically,
- replaces buyers/sellers matching with a pricing formula.

No brokers, no intermediaries.

Q9. What is a constant-product AMM?

A common AMM pricing rule:

$$x \times y = k$$

Where:

- x = credit reserve,
- y = stablecoin reserve,
- k = constant.

Buying one asset automatically adjusts price based on scarcity.

Q10. Why use AMMs for carbon credits?

AMMs:

- reduce transaction costs,
 - ensure continuous liquidity,
 - support retail PAYG use cases,
 - discourage hoarding by enabling easy exit.
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C. Carbon Market Problems & Concepts

Q11. What is hoarding in carbon credits?

Hoarding is buying credits and holding them indefinitely without retiring them.

Example:

A trader buys credits in 2025 and holds them until 2035, never offsetting emissions.

Problems:

- artificial scarcity,
 - price manipulation,
 - zero climate impact.
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Q12. How does this system prevent hoarding?

Through:

- expiry / decay rules,
- AMM liquidity pressure,
- PAYG micro-retirement,
- enforced PoO for claims.

Hoarding becomes economically irrational.

Q13. What is greenwashing?

Greenwashing is claiming environmental benefits without real impact.

Examples:

- claiming “carbon neutral” without retiring credits,
 - using low-quality or double-counted offsets.
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Q14. How does the system prevent greenwashing?

By enforcing:

- Proof-of-Integrity (PoI) before minting,
- Proof-of-Offset (PoO) at retirement,
- serialized, auditable receipts.

Claims become cryptographically verifiable, not marketing statements.

Q15. What is double counting?

Double counting occurs when the same emissions reduction is claimed more than once.

Example:

- a country claims reductions for national targets,
 - a company claims the same reduction for ESG reporting.
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Q16. How is double counting prevented?

By construction:

- serialization,
- bounded minting,
- exact serial burning,
- unique PoO issuance.

There is no valid path to claim the same serial twice.

D. MRV, Auditing & Data Integrity

Q17. What is MRV?

MRV = Measurement, Reporting, Verification

It ensures emissions reductions are:

- accurately measured,
 - consistently reported,
 - independently verified.
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Q18. Why keep MRV data off-chain?

Because raw MRV data is:

- large,
- sensitive,
- expensive to store on-chain.

Instead:

- raw data stays off-chain,
 - hashes + Merkle roots anchor integrity on-chain.
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Q19. What is a Merkle root and why use it?

A Merkle root is a single hash representing many data points.

Benefits:

- tamper evidence,
 - efficient verification,
 - constant-time integrity checks.
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Q20. What is a Zero-Knowledge Proof (ZKP)?

A ZKP proves:

“Valid data satisfies condition X” without revealing the data itself.

Q21. Why use ZKPs in carbon credits?

To:

- protect proprietary industrial data,
 - preserve confidentiality,
 - still enforce trustless verification.
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Q22. Why not rely on AI-only MRV verification?

AI:

- detects anomalies,

- does not prove correctness.

ZKPs:

- provide mathematical guarantees.

This system uses AI for detection, ZKPs for enforcement.

E. Two-Proof Model (Core Contribution)

Q23. What is Proof-of-Integrity (PoI)?

PoI proves:

“This credit is legitimate and allowed to exist.”

It is:

- generated by accredited VVBs,
 - ZKP-verified,
 - mandatory before minting.
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Q24. What is Proof-of-Offset (PoO)?

PoO proves:

“This credit has been permanently used.”

It is:

- generated at retirement,
 - non-transferable,
 - the only valid offset claim.
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Q25. Why separate PoI and PoO?

Because:

- legitimacy $\neq$ consumption,
- issuance $\neq$ settlement.

This separation removes:

- zombie credits,

- lifecycle ambiguity,
 - double claims.
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Q26. Why is PoO a separate object instead of a flag?

A flag:

- disappears with state changes,
- cannot be exported,
- lacks evidentiary value.

PoO:

- is a cryptographic receipt,
 - is portable,
 - persists forever.
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Q27. Why is PoO non-transferable?

Because offset claims:

- represent responsibility,
- must be unique,
- must not be traded.

Transferable PoO would reintroduce double-claim risk.

F. Registry, Governance & Trust

Q28. Why is the registry superior to markets?

Markets:

- discover price.

Registry:

- controls existence,
- enforces lifecycle,
- guarantees integrity.

This prevents market manipulation of validity.

Q29. Isn't the registry centralized?

No discretionary power exists.

All actions are:

- rule-based,
- cryptographically enforced,
- publicly auditable.

Power is constrained by code, not operators.

Q30. What happens if a VVB misbehaves?

Mitigations:

- whitelisting,
 - on-chain revocation,
 - PoI invalidation,
 - audit trails and slashing (optional).
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Q31. Can governments or regulators verify this system?

Yes.

They can:

- run read-only nodes,
 - verify PoI and PoO,
 - audit claims without accessing raw data.
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Q32. How are methodologies updated over time?

Methodologies are:

- versioned,
- whitelisted,
- locked per project at registration.

No retroactive rule changes.

G. Lifecycle, Risk & System Behavior

Q33. Why enforce a strict lifecycle?

Without it:

- credits float indefinitely,
- responsibility blurs,
- claims lose meaning.

Lifecycle enforced:

Existence → Trade → Consumption → Settlement

Q34. What happens when credits expire?

Expired credits:

- cannot be traded,
- cannot be retired,
- lose economic value.

This enforces timely use.

Q35. Why allow partial (fractional) retirement?

Because emissions occur continuously.

Enables:

- PAYG,
 - IoT-linked offsets,
 - real-time settlement.
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Q36. What if MRV data is challenged after minting?

Serialization allows:

- targeted invalidation,
 - freezing affected batches,
 - avoiding system-wide rollback.
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Q37. What if sensors fail or produce noise?

Handled via:

- aggregation windows,
- uncertainty buffers,
- conservative issuance bounds.

Over-crediting is structurally prevented.

Q38. Can credits be bridged across blockchains?

Yes, via wrapped credits that preserve:

- serialization,
- PoI/PoO references,
- registry authority.

Ownership moves, authority does not.

Q39. How does the system scale to billions of micro-offsets?

Via:

- batch serialization,
 - range-based PoO,
 - constant-size ZK proofs,
 - L2 settlement.
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H. Big-Picture & Conceptual Questions

Q40. Is this a carbon market or a carbon settlement system?

It is a carbon settlement protocol.

Markets are a side effect. Settlement is the purpose.

Q41. How is this different from existing systems?

Existing systems:

- track credits.

This system:

- cryptographically settles carbon liabilities.
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Q42. What happens if the platform disappears?

PoI and PoO:

- are on-chain,
- independently verifiable,
- persist without the operator.

Claims remain valid forever.

Q43. Who benefits most from this system?

- Corporates (verifiable ESG),
 - Retail users (PAYG),
 - Regulators (audit-grade trails),
 - Developers (liquidity),
 - Climate (real offsets).
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Q44. What is the strongest claim you can make?

It is impossible, not just unlikely, to mint, trade, or claim carbon credits dishonestly within this protocol.

Q45. One-sentence technical summary?

A registry-first, proof-driven carbon settlement protocol that cryptographically separates credit legitimacy from consumption using ZK-enforced issuance and serialized retirement.